
Metacognitive Sensitivity for Test-Time Dynamic Model Selection

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Abstract

1 A key aspect of human cognition is metacognition - the ability to assess one's
2 own knowledge and judgment reliability. While deep learning models can express
3 confidence in their predictions, they often suffer from poor calibration, a cognitive
4 bias where expressed confidence does not reflect true competence. Do models
5 truly know what they know? Drawing from human cognitive science, we propose
6 a new framework for evaluating and leveraging AI metacognition. We introduce
7 *meta-d'*, a psychologically-grounded measure of metacognitive sensitivity, to char-
8 acterise how reliably a model's confidence predicts its own accuracy. We then use
9 this dynamic sensitivity score as context for a bandit-based arbiter that performs
10 test-time model selection, learning which of several expert models to trust for a
11 given task. Our experiments across multiple datasets and deep learning model
12 combinations (including CNNs and VLMs) demonstrate that this metacognitive
13 approach improves joint-inference accuracy over constituent models. This work
14 provides a novel behavioural account of AI models, recasting ensemble selection
15 as a problem of evaluating both short-term signals (confidence prediction scores)
16 and medium-term traits (metacognitive sensitivity).

1 Introduction

18 Deep learning research is characterised by increasing specialisation. Convolutional Neural Networks
19 (CNNs) are widely used for perceptual tasks such as image recognition; Transformers and Large
20 Language Models (LLMs) dominate natural language processing; and Vision-Language Models
21 (VLMs) integrate across modalities Hu & Frank (2024); Gadetsky et al. (2025); McCoy et al. (2024);
22 Steyvers et al. (2025). This specialisation improves performance on benchmark tasks but also reflects
23 the "No Free Lunch" theorem: no single architecture is optimal across all problems Bigelow et al.
24 (2023). As a result, systems rely on multiple specialised models, raising the arbitration problem
25 - determining which model should be used for a given input. This is the focus of dynamic model
26 selection, which develops methods for assigning tasks to the most appropriate model.

27 A central obstacle in this area is the unreliability of model confidence. Deep neural networks produce
28 probabilistic outputs that are often miscalibrated, meaning confidence does not align with true
29 prediction accuracy. Research has shown that this miscalibration arises from the same architectural
30 and training practices that enable high accuracy Guo et al. (2017); Geirhos et al. (2018); Steyvers &
31 Peters (2025); Song et al. (2025a).

32 Cognitive science has long studied the analogous human ability, termed *metacognition*, providing
33 theoretical and mathematical tools for assessing how well an agent can evaluate its own knowl-
34 edge Rouault et al. (2019); Fleming (2021); Lee et al. (2025). Concepts such as metacognitive
35 sensitivity are increasingly viewed as necessary for developing AI systems that can self-monitor,
36 detect errors, and adjust behaviour based on uncertainty Yedetore et al. (2023); Kurvers et al. (2023);

37 Kuribayashi et al. (2025); Nguyen et al. (2025). Recent work has applied the *meta-d* framework to
 38 AI, introducing "AI metacognitive sensitivity" as a performance dimension. Findings suggest that in
 39 human-AI collaboration, an AI with lower accuracy but higher metacognitive sensitivity can be a
 40 more effective partner, as its confidence signals are more reliable for guiding when its advice should
 41 be followed.

42 While the literature has started to recognise the importance of AI metacognition as a diagnostic
 43 tool Gandhi et al. (2024); Ivanova et al. (2024); Gandhi et al. (2025); Song et al. (2025c); Murthy
 44 et al. (2025); Song et al. (2025b); Hu et al. (2025), the work reviewed here represents a critical next
 45 step: operationalising this concept as a functional component within an adaptive algorithm for a
 46 concrete engineering problem like dynamic model selection at test-time. It moves the idea of AI
 47 metacognition from a desirable property to be measured to a dynamic signal to be actively leveraged.

48 In this work, we embed metacognitive sensitivity into a bandit-based framework for dynamic model
 49 selection, demonstrating accuracy gains compared to individual constituent models. Beyond these
 50 improvements, our results suggest that leveraging metacognition as a functional signal can enable
 51 ensemble systems that adapt more flexibly and reliably.

52 2 Related Work

53 Given a pool of expert models, a system must decide how to aggregate their outputs to arrive at a
 54 final prediction. The paradigms for combining and selecting models are summarised as follows:

55 **Static Ensembles:** combine predictions from all models using fixed rules, such as majority voting
 56 for classification or averaging for regression Ganaie et al. (2022). Static ensembles reduce variance
 57 but are non-adaptive, treating all models as equally competent and using all of them for every input,
 58 which can be computationally inefficient when models have localised competence Ju et al. (2018); Su
 59 et al. (2021); Jiang et al. (2024).

60 **Dynamic Ensembles Selection:** selects a model or subset of models for each test instance, based
 61 on the idea that different classifiers perform best in different regions of the feature space Cruz et al.
 62 (2015); Liu et al. (2023); Piwko et al. (2025); Vasheghani & Sharifi (2025).

63 **Mixture of Experts (MoE):** consists of expert networks and a gating network Chen et al. (2022).
 64 The gating network assigns weights to experts for each input, and the final prediction is a weighted
 65 sum of expert outputs. Unlike Dynamic Selection, which uses pre-trained models with an external
 66 selection mechanism, MoE is trained end-to-end, with experts and gating network co-adapting during
 67 training Shazeer et al. (2017); Li et al. (2024, 2025); Mu & Lin (2025).

68 The shift from static ensembles to dynamic selection, and MoE methods underscores the need for
 69 adaptive mechanisms. Existing methods rely on fixed heuristics like local accuracy rather than learned
 70 policies. Reinforcement learning, particularly contextual bandits, offers a framework for adaptive
 71 selection by balancing exploration and exploitation across models. However, current approaches lack
 72 metacognition-aware feedback. They either depend on miscalibrated raw confidence or on noisy,
 73 static heuristics such as local accuracy. None leverage measures of metacognitive sensitivity to guide
 74 adaptive selection.

75 3 Problem Formulation

76 The central problem is to perform **test-time dynamic model selection** for an image classification task.
 77 Given a pair of pre-trained models, $M = \{M_A, M_B\}$, and a sequence of images, $D = \{x_1, \dots, x_N\}$,
 78 the objective is to create a framework that, for each image x_t , selects the model whose prediction
 79 is most likely to be correct. Since models often vary in performance across domains, particularly
 80 on out-of-distribution data, a static choice is suboptimal. We propose a dynamic selection agent
 81 that chooses between models per sample, informed by both immediate confidence and a cognitively
 82 inspired, dynamically updated measure of **metacognitive sensitivity** Fleming & Daw (2017).

83 The goal is to learn a selection policy π that maximises the cumulative reward, which is equivalent to
 84 maximising the total classification accuracy of the framework over the dataset:

$$\max_{\pi} \sum_{t=1}^N R_t = \max_{\pi} \sum_{t=1}^N \mathbb{I}(\hat{y}_{a_t, t} = y_t) \quad (1)$$

85 where $a_t = \pi(s_t)$ is the model selected at time t based on context s_t , $\hat{y}_{a_t,t}$ is its prediction, and y_t is
 86 the ground truth label.

87 4 Dynamic Model Selection Framework

Algorithm 1 Test-Time Dynamic Model Selection

```

1: Require: Model set  $M = \{M_A, M_B\}$ , Dataset  $D = \{x_1, \dots, x_N\}$ , Burn-in size  $B = 100$ ,
   Window size  $W = 100$ , Update frequency  $F = 50$ .
2: Initialise: Contextual Bandit (LinUCB or LinTS).
3: Initialise: Total Reward  $R_{total} \leftarrow 0$ .

   ▷ Burn-in Phase
4: Collect performance data (confidence, reward)  $H_k = \{(c_{k,i}, r_{k,i})\}_{i=1}^B$  for each model  $M_k \in M$ 
   on the first  $B$  trials.
5: for each model  $M_k \in M$  do
6:    $\mu_{k,B} \leftarrow \text{meta-d}'(H_k)$  ▷ Compute initial metacognitive score
7: end for

   ▷ Dynamic Selection Phase
8: for  $t = B + 1$  to  $N$  do
9:   for each model  $M_k \in M$  do ▷ Context Formulation
10:    Get confidence  $c_{k,t}$  on image  $x_t$ .
11:    if  $(t - B) \pmod{F} = 1$  then
12:      Collect past performance  $H_k \leftarrow \{(c_{k,i}, r_{k,i})\}_{i=t-W}^{t-1}$ .
13:      Update metacognitive score:  $\mu_{k,t} \leftarrow \text{meta-d}'(H_k)$ .
14:    else
15:      Keep previous score:  $\mu_{k,t} \leftarrow \mu_{k,t-1}$ .
16:    end if
17:  end for
18:  Construct context vector
      
$$s_t = [c_{A,t}, \mu_{A,t}, c_{B,t}, \mu_{B,t}] \quad (2)$$

      ▷ Bandit Action & Reward
19:  Select model  $a_t$  using bandit policy on  $s_t$ .
20:   $a_t \leftarrow \arg \max_{k \in \{A,B\}} \pi_t(s_t, k)$  ▷  $\pi_t(s_t, k)$ : a specific bandit selection in Appendix
21:  Get prediction  $\hat{y}_{a_t,t}$  from model  $M_{a_t}$ .
22:  Observe true label  $y_t$  and calculate reward  $R_t = \mathbb{I}(\hat{y}_{a_t,t} = y_t)$ .
23:   $R_{total} \leftarrow R_{total} + R_t$ .
24:  Update bandit policy with  $(s_t, a_t, R_t)$ . ▷ Learning
25: end for
26: return Overall Accuracy:  $R_{total}/(N - B)$ .
```

88 We propose a framework in which a contextual bandit serves as the dynamic selection agent. At
 89 each time step t , the bandit observes a context vector from two candidate models, selects one to
 90 act (the “arm”), receives a reward based on prediction correctness, and updates its policy. For each
 91 input x_t , models M_A and M_B produce predictions and confidence scores, which are combined into a
 92 4-dimensional context vector as shown in Equation. 2.

93 The **short-term signal** $(c_{k,t})$ is the model’s raw *confidence* on the current image x_t , taken as the
 94 maximum value of its softmax output probability; and the **metacognitive sensitivity** $(\mu_{k,t})$ is the
 95 score representing the model’s recent historical ability to know when it knows. This is a more stable,
 96 medium-term trait, which was operationalised using the **meta-d’** values.

97 Sensitivity scores are initialised using the first 100 trials and updated every 50 trials via a 100-trial
 98 sliding window. To address the computational demands of hierarchical Bayesian inference, we
 99 developed a GPU-parallelised package for efficient *meta-d’* estimation. For learning and decision-
 100 making, we employ the contextual bandit algorithms (Further bandit algorithmic details are provided
 101 in the Appendix).

5 Results

We evaluate our framework using four diverse pre-trained models: AlexNet, GoogleNet (classical CNN), EfficientNet (efficiently scaled CNN), and Vision Transformer (ViT) (transformer-based model). This selection captures both convolutional and attention-based models, enabling the assessment of the adaptability of our framework across fundamentally different design choices. We evaluate accuracy at three checkpoints corresponding to the number of trials indicates how many images have been processed up to that point.

Table 1: Accuracy of the best individual image models and the joint framework on the CIFAR10.

Model Pair	300 trials		700 trials		1000 trials	
	Model	Comb.	Model	Comb.	Model	Comb.
AlexNet-ViT	62.4	69.5 (+7.1%)	64.8	66.2 (+1.4%)	62.4	65.9 (+3.5%)
AlexNet-GoogleNet	62.7	70.6 (+7.9%)	57.7	57.5 (-0.2%)	56.8	58.4 (+1.6%)
EfficientNet-ViT	67.7	75.9 (+8.2%)	66.4	68.0 (+1.6%)	66.4	67.8 (+1.4%)
EfficientNet-GoogleNet	54.8	59.0 (+4.8%)	53.6	55.8 (+2.2%)	54.8	57.3 (+2.5%)

Table 1 shows that the joint framework’s accuracy is significantly improved in the early trials and then stabilises at 1.4%–3.5% higher than that of the individual models. At certain points, however, the joint framework underperforms relative to the best individual model in the pair. This effect can be attributed to correlated errors: performance is limited when both models misclassify the same sample, a pattern more common among architecturally similar models that share inductive biases from pre-training. For example, in one pairing (Figure. 1), the metacognitive sensitivity of AlexNet dropped (at trial 700), but the bandit quickly adapted by shifting selection toward GoogleNet. (Learning dynamics graphs are detailed in the Appendix A.)

Despite these challenges, architectural diversity enhances complementarity. Heterogeneous pairings, such as CNN-Transformer models, exhibit fewer correlated errors and achieve higher accuracy. This highlights the importance of inductive-bias diversity for effective dynamic model selection.

Table 2: Performance of the best individual VLM and the joint framework on the CIFAR10 - PACS.

Model Pair	1500 trials		2500 trials		4000 trials	
	Model	Comb.	Model	Comb.	Model	Comb.
MetaCLIP-SigLIP	98.7	99.0 (+0.3%)	98.7	98.6 (0.0%)	98.4	98.5 (+0.1%)
CLIP-ALIGN	94.2	96.0 (+1.8%)	94.8	96.2 (+1.6%)	94.8	95.8 (+1.0%)

To further probe the framework’s robustness, we evaluated it under domain shift using Vision-Language Models (CLIP, ALIGN, SigLIP MetaCLIP). To process the OOD bias in VLMs evaluation, we augment PACS Xu et al. (2019) with CIFAR-10. PACS exhibits a 1:5 imbalance between photographic and non-photographic styles, which may skew the results. Adding CIFAR-10 introduces additional photographic diversity, yielding a more balanced and robust combined dataset for benchmarking. Table 2 shows that while individual model performance declines under more demanding conditions, the framework leverages complementary strengths across models. This results in an initial accuracy improvement of 0.3%–1.8%, though because VLMs are already so accurate on their own, the gains are noticeable but modest compared to what had been seen with image models. However these potential suggest that metacognition-aware selection holds promise for addressing challenging, test-time distributional shifts in prediction tasks.

6 Conclusion

In this work, we presented a metacognition-inspired framework for dynamic model selection, introducing *meta-d* as a measure of how reliably a model’s confidence predicts its accuracy. Our experiments show that embedding metacognitive sensitivity in a bandit-based framework for dynamic model selection yields accuracy gains compared to individual constituent models. Future directions include extending the framework to large language model ensembles and exploring richer reinforcement learning strategies for selection. Overall, this work provides a first step toward incorporating metacognition-inspired feedback into adaptive model selection, with the goal of creating more reliable and interpretable ensemble systems.

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249 A Appendix

250 A.1 Metacognitive Sensitivity (*meta-d*)

251 To implement the metacognitive sensitivity score μ_k , we adopt the ‘meta-d’ framework developed by
 252 Fleming & Lau and detailed in Fleming & Daw (2017). Metacognition refers to the ability to monitor
 253 one’s own cognitive processes, and in this context, how well a model’s confidence relates to its
 254 actual success. *Meta-d* is a metric derived from **Signal Detection Theory (SDT)** by Macmillan that
 255 quantifies this ability. Its primary advantage is that it provides a measure of metacognitive sensitivity
 256 that is independent of the model’s task performance and overall confidence bias. We calculate
 257 *meta-d* by fitting a hierarchical Bayesian model to the observed distributions of confidence ratings
 258 for correct and incorrect trials Fleming (2017). For our framework, the score $\mu_{k,t}$ for a model M_k at
 259 time t is its calculated meta-d value.

260 A.2 Dynamic Update Mechanism

261 A key feature of our framework is that the metacognitive sensitivity score, μ_k , is not static but
 262 dynamically updated to reflect recent performance.

- 263 1. **Initialisation:** A burn-in set of the first 100 trials is used to compute the initial scores, $\mu_{A,0}$
 264 and $\mu_{B,0}$.
- 265 2. **Sliding Window Update:** During the selection process, the *meta-d* score for each model is
 266 recalculated every 50 trials using a **sliding window** of the 100 most recent trials.

267 This updating mechanism enables the framework to adapt to non-stationarity in model performance,
 268 such as when the underlying data distribution shifts.

269 A.3 Contextual Bandit Selection

270 The learning and decision-making core of our framework is a contextual bandit. We adopt two well-
 271 established algorithms: Algorithm. 2: **LinUCB** (Linear Upper Confidence Bound) and Algorithm. 3:
 272 **LinTS** (Linear Thompson Sampling). The bandit’s interaction at each step t is as follows:

- 273 1. **Observe Context:** Receives the context vector s_t .
- 274 2. **Select Action:** The algorithm’s policy selects an action $a_t \in \{A, B\}$.
- 275 3. **Receive Reward:** The framework executes the chosen model M_{a_t} . The reward R_t is 1 if
 276 the model’s prediction is correct, and 0 otherwise.
- 277 4. **Update Policy:** The bandit uses the tuple (s_t, a_t, R_t) to update its internal model, improving
 278 its policy for future decisions.

Algorithm 2 Linear Contextual Upper Confidence Bound

Require: Number of arms K , observe context s_t with dimension d , exploration parameter α

- 1: Initialise $A_a \leftarrow I_d$ and $b_a \leftarrow \mathbf{0}_d$ for all $k \in \{A, B\}$
 - 2: **for** each round $t = 1, 2, \dots$ **do**
 - 3: Observe context vector $s_t \in \mathbb{R}^d$
 - 4: **for** each arm $k \in \{A, B\}$ **do**
 - 5: Compute $\hat{\theta}_k \leftarrow A_k^{-1} b_k$
 - 6: Compute $\pi_t(s_t, k) \leftarrow \hat{\theta}_k^\top s_t + \alpha \sqrt{s_t^\top A_k^{-1} s_t}$
 - 7: **end for**
 - 8: Choose arm $k_t \leftarrow \arg \max_{k \in \{A, B\}} \pi_t(s_t, k)$
 - 9: Observe reward r_t
 - 10: Update: $A_{k_t} \leftarrow A_{k_t} + s_t s_t^\top$, $b_{k_t} \leftarrow b_{k_t} + r_t s_t$
 - 11: **end for**
-

Algorithm 3 Linear Contextual Thompson Sampling

Require: Number of arms K , observe context s_t with dimension d , prior variance parameter σ

- 1: Initialise $A_a \leftarrow I_d, b_a \leftarrow \mathbf{0}_d$ for all $k \in \{A, B\}$
 - 2: **for** each round $t = 1, 2, \dots$ **do**
 - 3: Observe context vector $s_t \in \mathbb{R}^d$
 - 4: **for** each arm $k \in \{A, B\}$ **do**
 - 5: Compute $A_k^{-1} \leftarrow (A_k + \epsilon I_d)^{-1}$
 - 6: Compute posterior mean: $\hat{\theta}_k \leftarrow A_k^{-1} b_k$
 - 7: Sample $\tilde{\theta}_k \sim \mathcal{N}(\hat{\theta}_k, \sigma^2 A_k^{-1})$
 - 8: Compute sampled reward: $\pi_t(s_t, k) \leftarrow \tilde{\theta}_k s_t^\top$
 - 9: **end for**
 - 10: Choose arm $k_t \leftarrow \arg \max_{k \in \{A, B\}} \pi_t(s_t, k) \rightarrow$ Observe reward r_t
 - 11: Update: $A_{k_t} \leftarrow A_{k_t} + s_t s_t^\top, \quad b_{k_t} \leftarrow b_{k_t} + r_t s_t$
 - 12: **end for**
-

279 **A.4 Learning Dynamics Graphs**

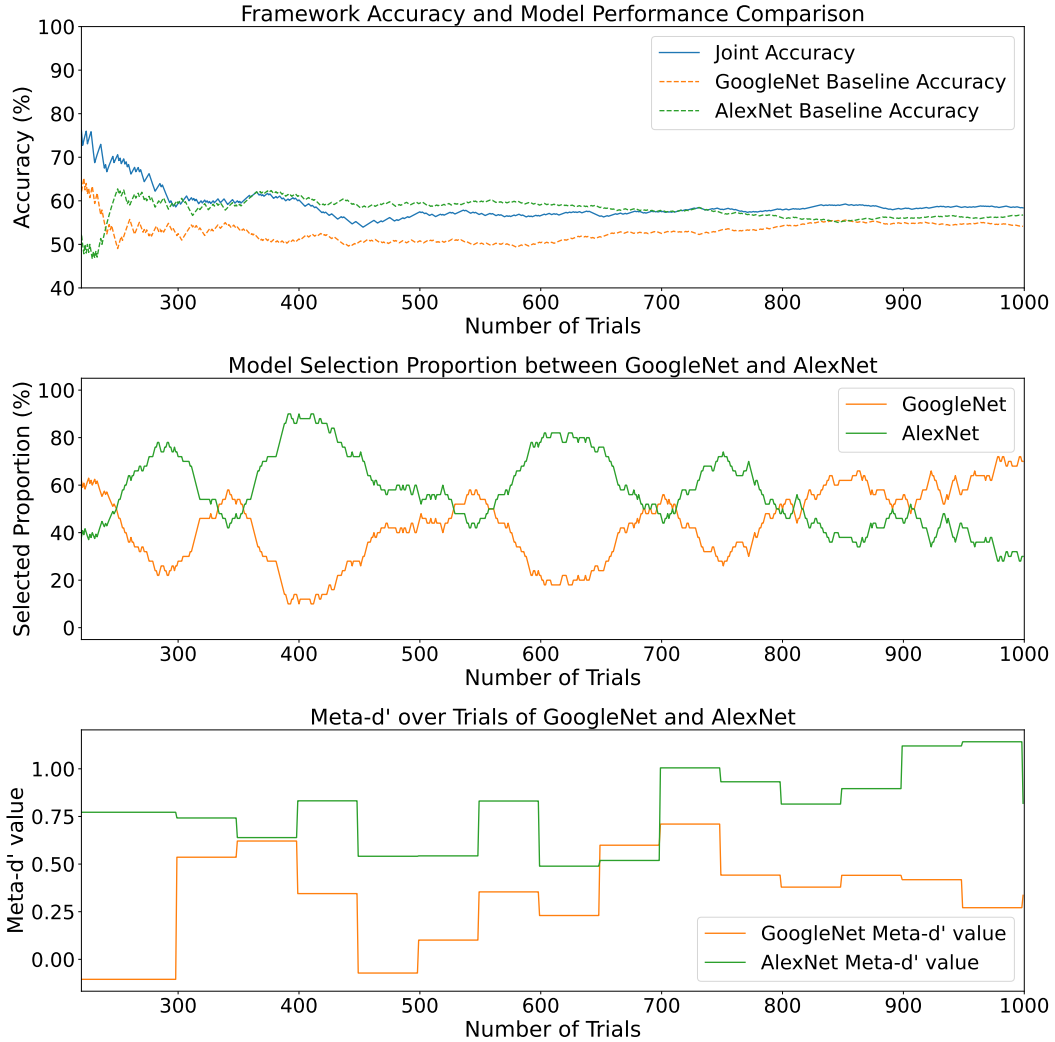


Figure 1: Figure of GoogleNet and AlexNet with Framework Accuracy (using LinTS with $\sigma = 0.5$)

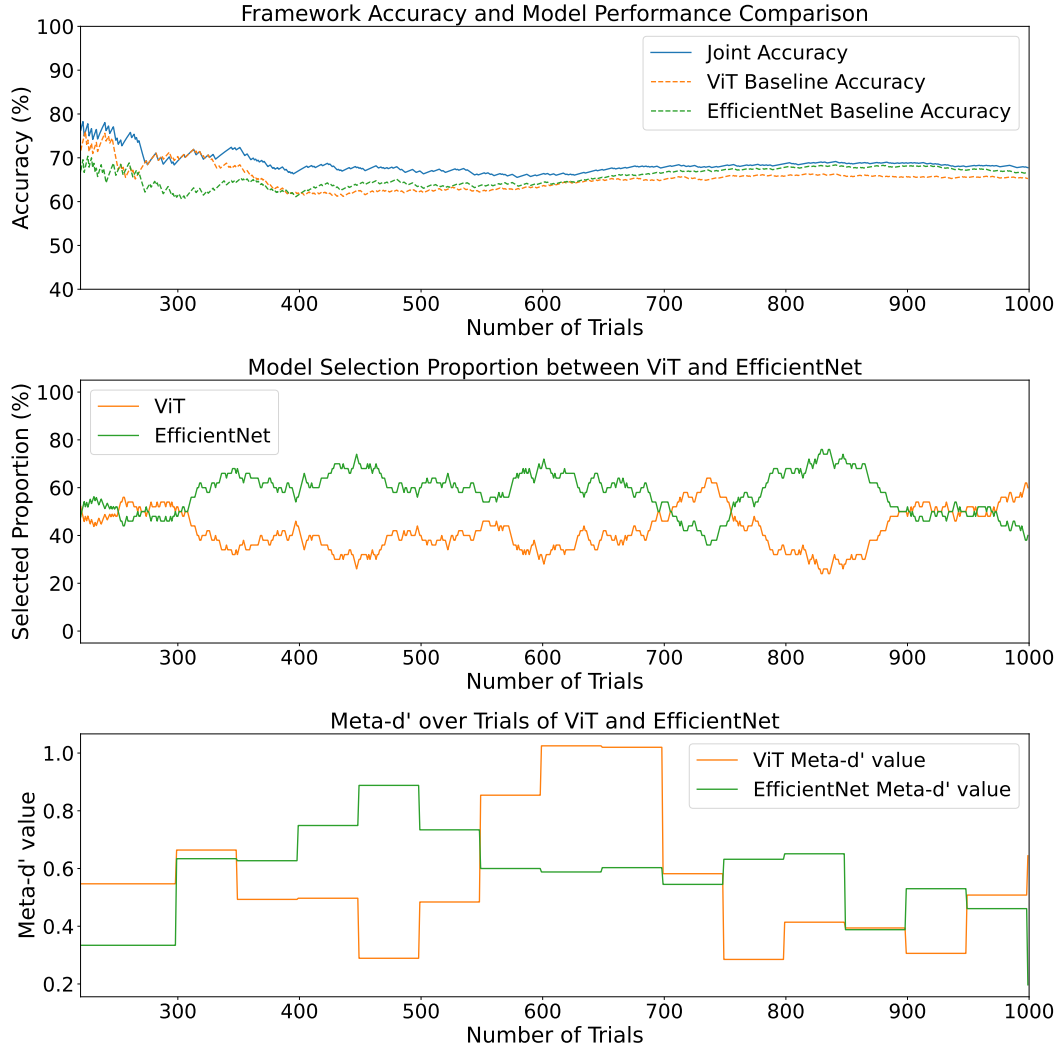


Figure 2: Figure of ViT and EfficientNet with Framework Accuracy (using LinTS with $\sigma = 1.0$)

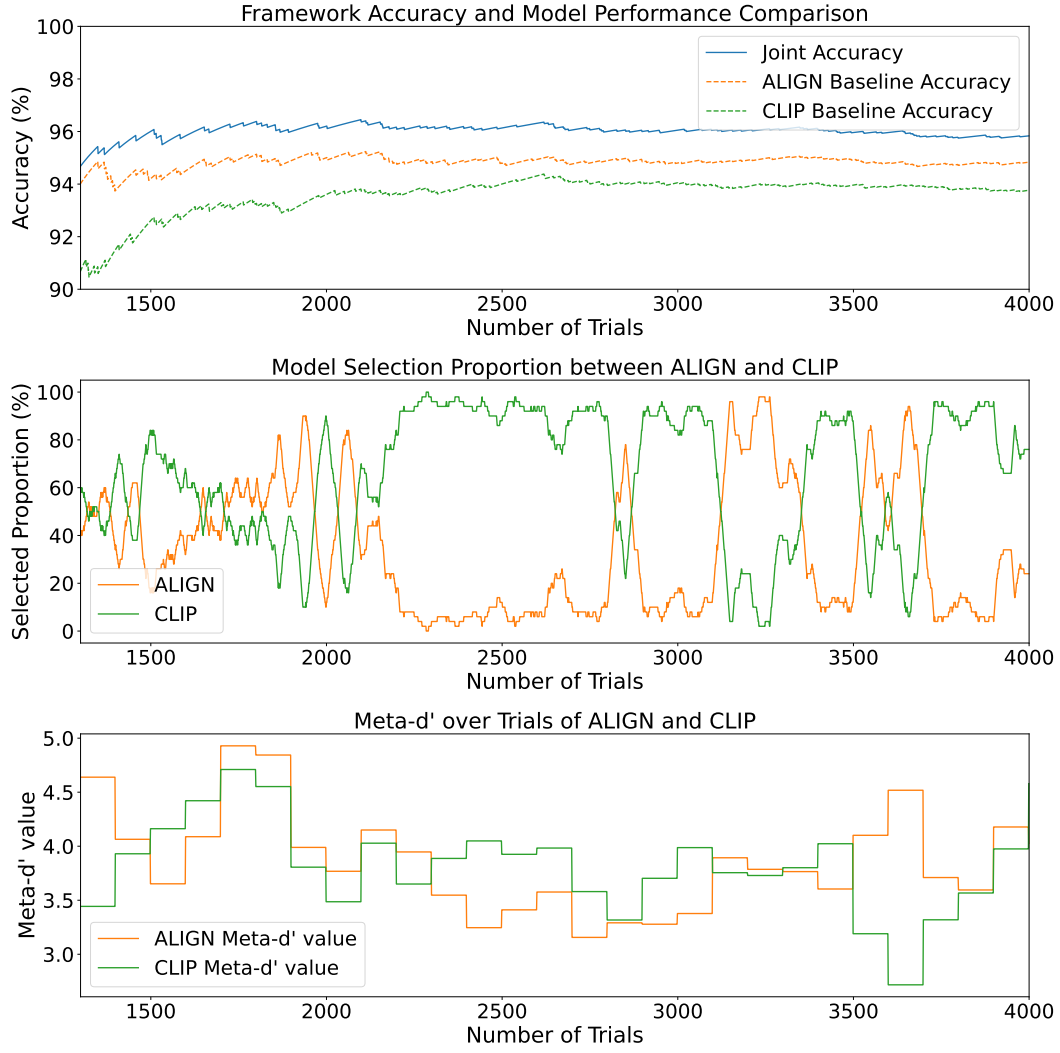


Figure 3: Figure of ALIGN and CLIP with Framework Accuracy (using LinUCB with $\alpha = 1.0$)

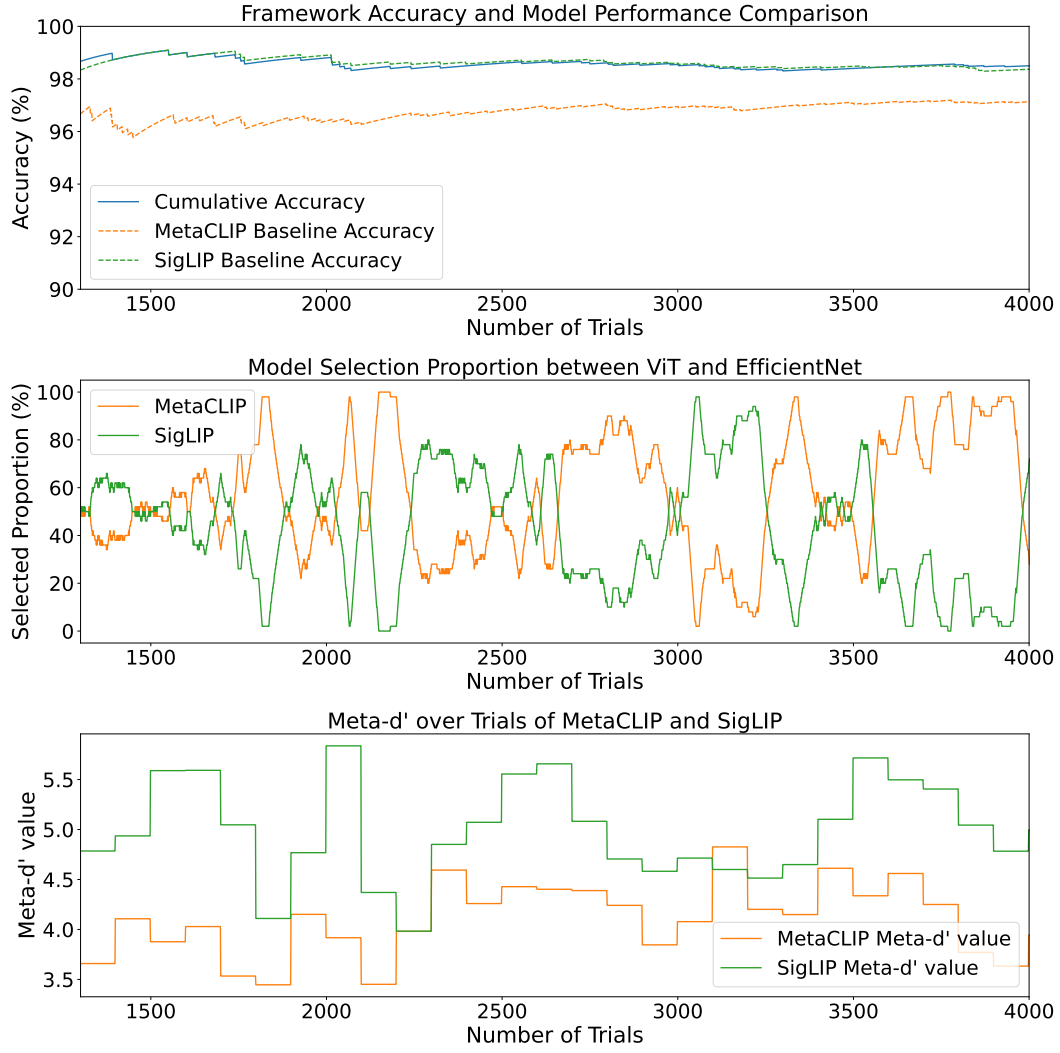


Figure 4: Figure of MetaCLIP and SigLIP with Framework Accuracy (using LinUCB with $\alpha = 0.5$)